BTP Progress Report — Part 3

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1 Scope of Part 3

Part 3 covers Tasks 8 to 11 of the implementation plan. Task 8 is the R descriptive analysis. Task 9 is the Makefile and the clean-clone rebuild check. Task 10 is the structure of the BTP-I report. Task 11 is the Beamer slides. Part 3 builds on the Git tag `part2`.

2 What was built

2.1 R descriptive functions and script

The file `src/R/lib/descriptives_fns.R` has three functions. `main_sample` keeps the rows with `in_main_sample` true. `sigma_cv` gives the coefficient of variation (standard deviation with $n-1$, divided by the mean) across the main-sample states for each period. `beta_convergence` fits an ordinary least squares regression of annualised growth on the logarithm of the initial level. The script `src/R/01_descriptives.R` reads `data/final/panel_quarterly.csv` and writes the outputs below. The tests are in `tests/R/test_descriptives.R`. The figures in `output/figures/` are:

- `adoption_trend_main.pdf`: median and interquartile range of ln(real value per person), by quarter.
- `sigma_convergence.pdf`: the coefficient of variation by quarter.
- `adoption_map_2018Q4.pdf`: state map for 2018Q4.
- `adoption_map_2026Q2.pdf`: state map for 2026Q2.

The tables in `output/tables/` are:

- `summary_stats.tex`: summary statistics.
- `beta_convergence.tex`: the two beta regressions (Section 3.1).
- `validation_npci.tex`: the PhonePe against NPCI correlations.

2.2 Makefile and clean-clone rebuild check

The `Makefile` has the targets `all`, `test`, `data`, `analysis` and `report`. `test` runs pytest and the R tests. `data` runs the Python modules from the raw-data check to the NPCI validation. `analysis` runs `src/R/01_descriptives.R`. `report` builds the report, the slides and the progress reports with latexmk. The rebuild check was run in a clean clone of the repository. The outputs of the clone were compared byte for byte with the outputs of the working tree. The compared files are the CSV files in `data/final`, `data/interim` and `data/manual`, and the files in `output/tables`. All compared files are identical. The figure PDFs are not compared. Each PDF has a creation date inside the file.

2.3 Report structure

The BTP-I report is `thesis/btp1/main.tex`. It has six chapters: Introduction, Literature Review, Data, Methodology, Preliminary results, and Plan for BTP-II. The student writes the prose of the chapters.

2.4 Slides

The slides are `thesis/btp1-slides/slides.tex`. The file has 14 frames. The PDF has 15 pages.

3 Numbers

3.1 Beta convergence

The sample is the main sample (30 states and union territories), 2018Q4 to 2026Q2. For each state, growth is $(\ln x_{2026Q2} - \ln x_{2018Q4})/7.5$. The window has 30 quarters, which is 7.5 years. The regression is growth on $\ln x_{2018Q4}$ with an intercept (OLS). The standard errors are the classical OLS standard errors. The slope for real value per person is positive (0.0177, standard

Table 1: Beta regressions, main sample, 2018Q4 to 2026Q2. Dependent variable: annualised growth of the variable.

	Real value per person	Transactions per person
Slope on ln(initial level)	0.0177	−0.0030
Standard error	0.0138	0.0128
R^2	0.056	0.002
N	30	30

error 0.0138). The slope for transactions per person is negative (−0.0030, standard error 0.0128). Neither slope is more than 1.3 standard errors from zero.

3.2 Coefficient of variation

The coefficient of variation (CV) is the sample standard deviation divided by the mean, across the 30 main-sample states. The CV of real value per person rose from 0.719 to 0.908. The CV of

Table 2: Coefficient of variation across the 30 main-sample states.

Variable	2018Q4	2026Q2
Transactions per person	0.854	0.865
Real value per person	0.719	0.908
Registered users per 1000 persons	0.622	0.379

transactions per person rose from 0.854 to 0.865. The CV of registered users per 1000 persons fell from 0.622 to 0.379.

3.3 Tests

The command `.venv/bin/pytest-q` reports 98 passed. The R command `testthat::test_dir("tests/R")` reports 0 failures, 0 warnings, 0 skipped and 3 passed expectations. The file `tests/R/test_descriptives.R` has 3 `test_that` blocks. These counts include the tests of Section 5.

3.4 Report length

The command `pdfinfothesis/btp1/main.pdf` reports 17 pages after `make report`.

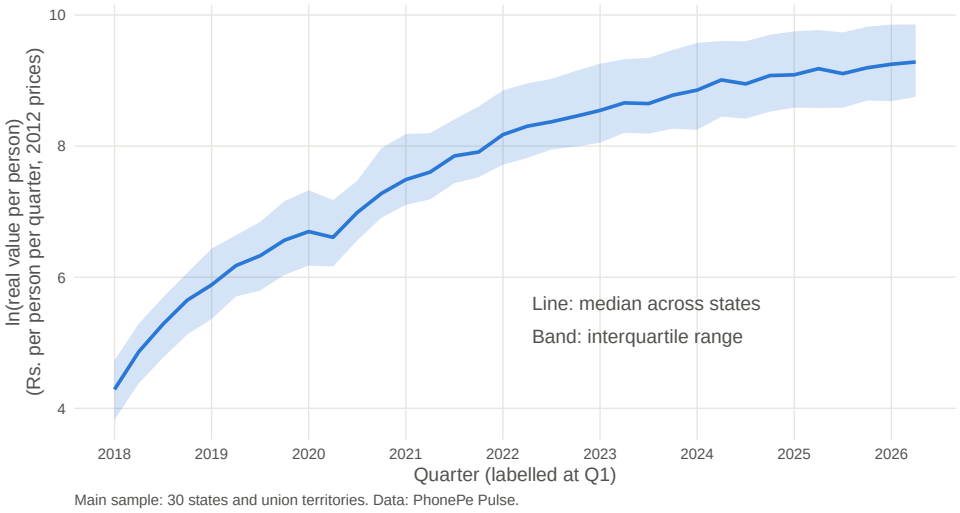


Figure 1: Median (line) and interquartile range (band) of the natural logarithm of real value per person (Rs. per person per quarter, 2012 prices), by quarter, 2018Q1 to 2026Q2. Sample: 30 main-sample states and union territories. Source: PhonePe Pulse [2]; deflator: linked state CPI [1].

4 Data notes for the thesis

In 2026Q2, the real value per person in Telangana is 62,328 Rs. (2012 prices). Telangana has 11.3% of the PhonePe transaction amount. The amount is Rs. 5.13×10^{12} of Rs. 4.55×10^{13}. The PhonePe file for 2026 quarter 2 and the panel give the same total and the same share. Telangana has 2.7% of the population of the 35 ids (38,664,500 persons).

5 Corrections after final review

A review of the full branch found the items below. Each item has one commit. The numbers in Sections 3 and 4 are the numbers after these corrections.

5.1 CPI link method

The old method linked base 2012 and base 2024 with an annual-mean factor, the ratio of the 2025 means:

$$f_s^{old} = \overline{CPI}^{2012}_{s,2025} / \overline{CPI}^{2024}_{s,2025}.$$

The new method uses the December 2025 factor:

$$f_s = CPI^{2012}_{s,2025\text{-}12} / CPI^{2024}_{s,2025\text{-}12}.$$

From 2026-01, the linked value is $CPI^{2024}_s \times f_s$. A state without December 2025 in both bases uses the latest 2025 month in both bases. All 34 state factors use December 2025. The new method gives 34 state factors. The minimum is 1.686 (Delhi). The median is 1.882. The maximum is 2.107 (Manipur). The factor for the all-India series is 1.902. The Part 2 report (`part2.pdf`, Git tag `part2`) shows the factor statistics of the old method: minimum 1.700 (Delhi), median 1.891, maximum 2.122 (Manipur), all-India 1.899. These statistics are superseded.

The step at the link, in percentage points, is

$$100 \times \Big(\frac{CPI^{link}_{s,\text{2026-01}}}{CPI^{link}_{s,\text{2025-12}}} - \frac{CPI^{2024}_{s,\text{2026-01}}}{CPI^{2024}_{s,\text{2025-12}}} \Big).$$

It is the difference between the monthly change of the linked series and the monthly change of base 2024. With the old factor, the largest step is -2.63 percentage points (Kerala). With the new factor, the largest absolute step over the 34 states is 2.2×10^{-14} percentage points. This is zero to floating-point precision. The test `test_real_link_keeps_base2024_dec_jan_change` in `tests/py/test_cpi.py` checks for each state that the two ratios in the formula differ by less than 10^{-9}.

The change of method changes `data/interim/cpi.csv`, `data/final/panel_quarterly.csv`, the four figures and the tables `beta_convergence.tex` and `summary_stats.tex`. The numbers in this report change as follows:

- Beta slope for real value per person: 0.0174 to 0.0177. R^2: 0.054 to 0.056. The standard error stays 0.0138.
- Beta slope for transactions per person: no change (-0.0030). This variable does not use the CPI.
- CV of real value per person in 2026Q2: 0.906 to 0.908. The 2018Q4 value stays 0.719.
- Real value per person in Telangana, 2026Q2: 61,997 Rs. to 62,328 Rs. (2012 prices).

The report length stays 17 pages.

5.2 Other corrections

- Merged state ids. The function `sum_merged` in `src/py/btpdata/states.py` sums the components of `jammu_kashmir` and `dnh_dd`. It finds the components in the whole source. If one key (year, quarter or month) does not have all components, it stops with an error. The population, PhonePe and NPCI modules use it. The data files after this change are byte-identical to the data files before it.
- Beta window. `beta_convergence` in `src/R/lib/descriptives_fns.R` calculates the window length in years from the period strings: (quarter index of the end − quarter index of the start)/4. The constant 7.5 is removed. For 2018Q4 to 2026Q2 the value is 7.5, so the table does not change.
- Raw-file check. `src/py/btpdata/check_raw.py` calculates the SHA-256 of each manual raw file and compares it with the `sha256` column of its manifest. If the values are different, it stops with a message that gives the file, the expected value and the actual value. The three manual raw files agree with their manifests.
- Report Figure 5.2 caption: the maps show all 35 units (28 states and 7 union territories as merged in the panel).
- Slides: the statement "PhonePe has about 46% of UPI volume" is removed. The slides now give the measured ratio below.

5.3 PhonePe Pulse count and NPCI volume

The ratio is the PhonePe Pulse transaction count of the 35 ids divided by the NPCI UPI volume. The NPCI volume is the sum of all rows without a district, with the unclassified row included.

- 2025Q2: 3.113×10^{10} PhonePe transactions, 5.497×10^{10} NPCI UPI transactions, ratio 0.566.
- 2026Q2: 3.866×10^{10} PhonePe transactions, 6.859×10^{10} NPCI UPI transactions, ratio 0.564.

The data in this repository do not show which transactions the Pulse count includes. Open question: does the Pulse count include only UPI transactions, or also other payment instruments? Until this question has an answer, this report does not use the ratio as the PhonePe share of UPI volume.

6 Open items

- Task 0: supervisor approval of the PhonePe proxy. Department rules for BTP-I.
- The student writes the prose of the report chapters.
- The file `references/refs.bib` has four entries with a `verify` field:
  - `pradhan2021tfsc`: "verify exact title and full author list".
  - `pradhan2021ael`: "verify exact title and second author's given name".
  - `reddy2024icrier`: "year not shown on page, verify".
  - `singh2021eap`: "verify authors' given names".
- The coverage of the PhonePe Pulse transaction count (Section 5).
- The open items of Part 2 stay open.

7 Next

BTP-II (Section 6, "After BTP-I", of the design specification) has these parts: financial inclusion data and indices, Steps 1 to 3 of the analysis, and the thesis. It also has a check with a district panel.

References

[1] Ministry of Statistics and Programme Implementation. Consumer price index, state-wise monthly, combined, general index: base 2012 and base 2024, eSankhyiki API. `https://esankhyiki.mospi.gov.in/macroindicators?product=cpi`, 2026. accessed 1 November 2026.

[2] PhonePe. PhonePe Pulse data repository. `https://github.com/PhonePe/pulse`, commit 257bf38e3095020f9f516c4c72c10bdc348781b1, 2026. CDLA-Permissive-2.0; data 2018Q1–2026Q2; accessed 1 November 2026.